1. Core model: City selection + vehicle deployment + multiple penalties

$$\min_{y,q,O,U} \quad \underbrace{\sum_c F_c y_c}_{\text{Entry costs}} + \underbrace{\sum_c K_c q_c}_{\text{Vehicle costs}} + \frac{1}{|S|} \sum_s \sum_c \left[\underbrace{c_c^+ O_c^s}_{\text{Remaining capacity}} + \underbrace{c_c^- U_c^s}_{\text{Foregone contribution margin}} \right].$$

In c^+, c^-, $+, -$ distinguish excess and shortage, rather than the signs of costs. All demand, throughput, and costs use **the same evaluation period.** ※ I suppose that setting the evaluation period is needed, and the model determines the city/vehicle deployment to apply during that period.

2. Constraints

$$\begin{aligned} 0 \le q_c \le \bar{q}_c y_c \qquad & \forall c, \\ \sum_c q_c \le Q, \qquad & \\ \kappa_c q_c - M_c p_c^s = O_c^s - U_c^s \qquad & \forall c, s, \\ y_c \in \{0,1\}, \quad q_c \in \mathbb{Z}_{\ge 0}, \qquad & \\ O_c^s, U_c^s \ge 0. \qquad & \end{aligned}$$

Constraint	Meaning
$0 \le q_c \le \bar{q}_c y_c$	Zero vehicles if the city is not entered; up to the city-specific limit if it is entered
$\sum_c q_c \le Q$	Total vehicle limit
$\kappa_c q_c - M_c p_c^s = O_c^s - U_c^s$	The difference between service capacity and robotaxi demand equals unused capacity minus unmet demand.

3. Explanations

A. Construct two types of demand

$$D_c^s = M_c p_c^s, \qquad D_c^{N,s} = M_c(1 - p_c^s)$$

Symbol	Meaning	Explanation
M_c	Total number of **potential travel opportunities across all modes**	Robotaxi demand + human-driven taxi demand + other demand
p_c^s	Probability of choosing robotaxi service for a travel opportunity in city c under the specified service conditions, before applying fleet-capacity constraints	Need to be specified with reference service conditions such as fares and waiting times
D_c^s	Robotaxi choice demand	Demand **before** applying vehicle shortages
$D_c^{N,s}$	Demand not choosing robotaxis	Human-driven taxis, other modes, giving up the trip, etc.

B. Constraints

1) Once demand D_c^s and service capacity $\kappa_c q_c$ are determined, the following two values are calculated based on the deployment outcome; they are not variables that need to be investigated separately. $(x)^+ = \max\{x, 0\}$. Both values are measured in trips.

$$\begin{aligned} O_c^s &= (\kappa_c q_c - D_c^s)^+ \quad \text{Remaining service capacity,} \\ U_c^s &= (D_c^s - \kappa_c q_c)^+ \quad \text{Unserved potential demand.} \end{aligned}$$

4. Notation used in the model

Symbol	Meaning	Type and unit
C	Set of potential candidate cities	Input set; cities
S	Set of demand scenarios	Input set; scenarios
y_c	1 if city c is entered, else 0	Decision variable; binary (0 or 1)
q_c	Number of vehicles deployed in city c	Decision variable; vehicles (integer)
κ_c	Number of trips one vehicle can serve during the period	Scale parameter; trips/vehicle over the evaluation period.
Q	Total number of vehicles available for deployment	Input parameter; vehicles
$\bar{q}_c$	Upper bound on vehicle deployment in each city	Input parameter; vehicles
F_c	Fixed cost of city entry and basic infrastructure	Input parameter; currency over the evaluation period
K_c	Per-vehicle preparation, deployment, and holding cost over the evaluation period	Input parameter; currency/vehicle over the evaluation period
c_c^+	Additional loss per unit of unused service capacity	Input parameter; currency/trip of unused capacity
c_c^-	Foregone contribution margin per unit of unserved potential demand	Input parameter; currency/unserved trip
M_c	Total number of **potential travel opportunities across all modes** in city c during one year. c	Input parameter; usage opportunities over the evaluation period. data or assumption
p_c^s	Probability of choosing robotaxi service for a travel opportunity in city c under the specified service conditions, before applying fleet-capacity constraints.	Scenario input; dimensionless probability in $[0, 1]$
D_c^s	Robotaxi choice demand	Derived input; trips over the evaluation period. $D_c^s = M_c p_c^s$
$D_c^{N,s}$	Demand not choosing robotaxis (human-driven taxis, other modes, giving up the trip, etc.)	Derived input; trips over the evaluation period. $D_c^{N,s} = M_c(1 - p_c^s)$
O_c^s	Unused service capacity	Outcome variable; trips over the evaluation period. $O_c^s = (\kappa_c q_c - D_c^s)^+$
U_c^s	Unserved robotaxi demand	Outcome variable; trips over the evaluation period. $U_c^s = (D_c^s - \kappa_c q_c)^+$

※ Demand-related variables consist of the inputs p_c^s, M_c and outcome variables calculated from scenario demand and deployment in the model. For the baseline model, M_c is fixed. ※ Outcome variables are not quantities to be predicted by collecting data for each one separately. They are outcomes calculated from demand and deployment.